Pharmaceutical Organic Chemistry - II (Course Code: BP301T)

Uploaded By: Priya Prajapati (B.Pharm 3rd Semester) | Official Course Curriculum | Peer Verified & 4.9

Unit 1: Benzene & Electrophilic Aromatic Substitution

- Kekulé Structure & Resonance Stability of Benzene ring with 6 delocalized < VÆV7G&öç0.
- Nitration Mechanism: Generation of Nitronium ion (NO2+) using H2SO4 + HNO3, forming Arenium ion intermediate.
- Friedel-Crafts Alkylation & Acylation: AlCl3 Lewis acid catalyzed carbocation attack and synthetic limitations.
- Substituent Effects: Ortho/Para directors (activating -OH, -NH2, -CH3) vs Meta directors (deactivating -NO2, -COOH, -SO3H).

Unit 2: Phenols & Aromatic Amines Structure-Activity Guide

- Acidity of Phenols: Resonance stabilization of Phenoxide ion compared to aliphatic alcohols.
- Synthesis & Quantitative Tests: Ferric Chloride test, Liebermann test, and Bromine water precipitation.
- Basicity of Aromatic Amines: Effect of electron-withdrawing vs donating substituents on pKa.